



Comprehensive Guide to the Nervous System (BIO101 Midterm Notes)

Animal Anatomy And Physiology (Fort Valley State University)



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Comprehensive Guide to the Nervous System

Introduction

The nervous system is a complex network that coordinates actions and sensory information by transmitting signals to and from different parts of the body. It is divided into two main parts: the Central Nervous System (CNS) and the Peripheral Nervous System (PNS).

Functions of the Nervous System

- **Receive Stimuli:** Sensory receptors collect information from the environment.
- **Analyze and Integrate Stimuli:** This data is processed and interpreted.
- **Respond to Stimuli:** Suitable responses are generated and executed.

Subdivisions of the Nervous System

Central Nervous System (CNS)

- **Brain:** The control center of the body, responsible for thought, memory, emotion, touch, motor skills, vision, breathing, temperature, and many other functions.
- **Spinal Cord:** Connects the brain and the peripheral nervous system, facilitating the transmission of neural signals.

Peripheral Nervous System (PNS)

- **Cranial Nerves:** Originate from the brain; primarily control the motor and sensory functions of the head and neck.

- **Spinal Nerves:** Emerge from the spinal cord; they facilitate communication between the CNS and the rest of the body.
- **Autonomic Nerves:** Regulate involuntary body functions, such as heart rate, digestion, and respiratory rate.

Autonomic Nervous System (ANS)

- **Sympathetic Nervous System (SNS):** Prepares the body for stressful or emergency situations—often referred to as the "fight-or-flight" response.
- **Parasympathetic Nervous System (PSNS):** Promotes a "rest-and-digest" state, conserving energy and facilitating digestion and repair.

Neuron Structure and Function

- **Cell Body:** Contains the nucleus and metabolic center.
- **Dendrites:** Carry impulses towards the cell body.
- **Axon:** Transmits impulses away from the cell body.
- **Myelin:** Insulating layer that increases the speed of impulse transmission.

Synapse and Neurotransmitters

- **Synapse:** Junction between two neurons or a neuron and a target cell.
- **Neurotransmitters:** Chemical messengers that facilitate communication across the synaptic cleft. Examples include acetylcholine, norepinephrine, epinephrine, dopamine, gamma-aminobutyric acid (GABA), and glycine.

Generating a Nerve Impulse

- **Threshold Stimulus:** Sufficient strength to trigger a nerve impulse.

- **All-or-Nothing Principle:** Once threshold is reached, a nerve impulse occurs completely.
- **Depolarization and Repolarization:** The electrical state change that transmits the impulse.

Brain Anatomy and Functions

- **Forebrain:** Includes the cerebrum, thalamus, and hypothalamus; involved in sensory and associative functions and connecting the endocrine and nervous systems.
- **Midbrain:** Acts as a relay station for auditory and visual information.
- **Hindbrain:** Comprises the cerebellum (balance and coordination), pons, and medulla oblongata (regulates vital body functions).

Protection Mechanisms of the CNS

- **Cranium:** Bony structure protecting the brain.
- **Meninges:** Three-layered protective tissue (dura mater, arachnoid mater, pia mater) around the brain and spinal cord.
- **Cerebrospinal Fluid (CSF):** Cushions the CNS and removes waste.
- **Blood-Brain Barrier:** Prevents harmful substances from reaching the brain while allowing essential nutrients to pass through.

Reflexes and Reflex Arcs

- **Reflexes:** Automatic, rapid responses to stimuli designed to protect the body and maintain homeostasis. Examples include withdrawal, anal, patellar, panniculus, and palpebral reflexes.
- **Reflex Arc:** Pathway that involves a sensory neuron sending a signal to the spinal cord where it connects to a motor neuron, prompting a response.

Peripheral Nervous System

- **Cranial Nerves:** 12 pairs that control sensory and motor functions.
- **Spinal Nerves:** Correspond to each segment of the spinal cord, facilitating sensory and motor pathways.
- **Plexuses:** Complex networks of nerves, such as the brachial plexus (forelimbs) and lumbosacral plexus (hindlimbs).

Conclusion

The nervous system's intricate structure and function enable it to control and coordinate various activities in the body efficiently. Its divisions and numerous components work in harmony to process information and respond appropriately to internal and external stimuli.

References:

- Content summarized from provided slides.